

Linux Syslog Security Monitoring & Log Analysis Report

Author

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Role

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Project Title

Linux Syslog Security Monitoring & Log Analysis Lab

Report Version

1.0

Assessment Date

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Executive Summary

This project focused on performing security monitoring and log analysis on a Linux-based system using the rsyslog logging framework and native Linux log sources.

The objective was to investigate system-generated events, authentication activity, failed login attempts, privileged account usage, and system service activity to demonstrate practical Security Operations Center (SOC) monitoring techniques.

Using Ubuntu Linux, rsyslog, syslog, and authentication logs, multiple security-relevant events were identified and analyzed. The investigation included authentication failures, root account access, sudo activity, user session monitoring, and log source summarization.

The project successfully demonstrated foundational SOC analyst skills including event correlation, threat detection, authentication auditing, log parsing, incident investigation, and MITRE ATT&CK mapping.

No evidence of system compromise or malicious persistence was identified during the assessment. All detected activities were generated during controlled lab testing.

Project Objectives

The primary objectives of this project were:

- Understand Linux logging architecture
 - Review rsyslog configuration
 - Analyze system and authentication logs
 - Detect authentication failures
 - Investigate privileged account activity
 - Monitor user sessions
 - Perform event correlation
 - Apply MITRE ATT&CK mapping
 - Develop practical SOC investigation skills
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Lab Environment

Component	Details
Operating System	Ubuntu 24.04 LTS
Logging Service	rsyslog
Primary Logs	syslog, auth.log
Analysis Tools	grep, awk, less, sort, uniq
Investigation Type	Security Monitoring & Log Analysis

Linux Logging Architecture

Linux systems generate operational and security events through multiple services and applications.

The rsyslog daemon collects, processes, and stores these events within the /var/log directory.

Key log sources investigated during this assessment included:

Syslog

General operating system and application activity.

Authentication Log (auth.log)

Authentication attempts, login activity, sudo usage, and privileged access events.

Systemd Events

System service startup and operational activity.

CRON Events

Scheduled task execution and automation activity.

Investigation Methodology

The following methodology was followed during the assessment:

Phase 1 – Logging Service Validation

Verified that rsyslog was installed and actively collecting system events.

Command

```
systemctl status rsyslog
```

Result:

rsyslog service confirmed operational.

Phase 2 – Configuration Analysis

Reviewed rsyslog configuration settings.

Configuration Components Verified

```
module(load="imuxsock")  
module(load="imklog")  
$IncludeConfig /etc/rsyslog.d/*.conf
```

Result:

Logging modules and configuration inheritance successfully validated.

Phase 3 – Log Collection Review

Investigated log sources located in:

```
/var/log/
```

Key log files reviewed:

- syslog
- auth.log
- kern.log
- dpkg.log

Result:

System logging functionality verified.

Phase 4 – Event Filtering

Log filtering was performed using grep.

Examples:

```
grep "systemd" /var/log/syslog
```

```
grep "CRON" /var/log/syslog
```

Result:

System service activity and scheduled task events successfully identified.

Phase 5 – Authentication Monitoring

Authentication events were analyzed using auth.log.

Examples:

```
grep "sudo" /var/log/auth.log
```

```
grep "session opened" /var/log/auth.log
```

```
grep "authentication failure" /var/log/auth.log
```

Result:

Successful identification of authentication activity, sudo events, and failed login attempts.

Security Findings

Finding 1 – Failed Authentication Attempts

Severity

Medium

Description

Multiple failed authentication attempts were identified within the authentication logs.

Evidence

```
authentication failure  
3 incorrect password attempts
```

Impact

Repeated failed authentication attempts may indicate:

- Password guessing
- Brute-force activity
- Unauthorized access attempts
- Weak credential attacks

MITRE ATT&CK Mapping

Technique ID	Technique
T1110	Brute Force
T1110.001	Password Guessing

Finding 2 – Root Account Access

Severity

Medium

Description

Root account activity was identified during authentication monitoring.

Evidence

session opened for user root

Impact

Root accounts possess unrestricted privileges and should be continuously monitored.

MITRE ATT&CK Mapping

Technique ID	Technique
T1078	Valid Accounts

Finding 3 – Administrative Activity

Severity

Informational

Description

Administrative actions executed through sudo were successfully recorded.

Evidence

```
sudo session opened  
sudo session closed
```

Impact

Provides accountability and audit visibility for privileged actions.

MITRE ATT&CK Mapping

Technique ID	Technique
T1098	Account Manipulation

Finding 4 – User Session Monitoring

Severity

Informational

Description

User login sessions and account activity were successfully tracked.

Evidence

```
session opened  
session closed
```

Impact

Provides visibility into user behavior and access patterns.

MITRE ATT&CK Mapping

Technique ID	Technique
T1033	System Owner/User Discovery

Incident Analysis

Event Summary

During the investigation, multiple authentication failures were observed followed by successful administrative access events.

Timeline

Event	Observation
Failed Authentication	Detected
Incorrect Password Attempts	3 Attempts
Sudo Activity	Detected
Root Access	Detected
User Sessions	Detected

Analyst Assessment

The observed activity was consistent with controlled laboratory testing.

No indicators of compromise were identified.

No persistence mechanisms were detected.

No malicious privilege escalation was observed.

MITRE ATT&CK Coverage

MITRE ID	Technique
T1110	Brute Force
T1110.001	Password Guessing
T1078	Valid Accounts
T1098	Account Manipulation
T1033	System Owner/User Discovery

Risk Assessment

Finding	Risk
Authentication Failures	Medium
Root Account Activity	Medium
Sudo Activity	Low
Session Monitoring	Low

Overall Risk Rating

Medium

The environment demonstrated realistic authentication-related security events that require monitoring in enterprise environments.

Remediation Recommendations

Authentication Security

- Enforce strong password policies
- Implement account lockout mechanisms
- Enable Multi-Factor Authentication (MFA)

Privileged Access Management

- Restrict direct root logins
- Monitor sudo activity
- Apply least privilege principles

Continuous Monitoring

- Deploy centralized logging
- Create alerts for authentication failures
- Monitor privileged account activity

Security Operations Improvements

Recommended platforms:

- Splunk
- ELK Stack

- Wazuh
 - Microsoft Sentinel
 - Google Cloud Logging
 - AWS CloudWatch Logs
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Skills Demonstrated

SOC Analyst Skills

- Log Analysis
- Security Monitoring
- Authentication Auditing
- Threat Detection
- Incident Investigation
- Event Correlation
- Risk Assessment
- MITRE ATT&CK Mapping

Cloud Security Skills

- Log Monitoring
 - Security Operations
 - Security Visibility
 - Event Analysis
 - Detection Engineering Fundamentals
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Conclusion

This project successfully demonstrated practical Linux log analysis and security monitoring techniques using rsyslog and native Linux logging sources.

Authentication failures, privileged account activity, session events, and system logs were analyzed using SOC investigation methodologies. Security findings were documented and mapped to MITRE ATT&CK techniques to align with modern security operations practices.

The project provided hands-on experience in security monitoring, threat detection, authentication auditing, and incident investigation, making it directly relevant to SOC Analyst and Cloud Security Engineer roles.